Qwen3.5-4B on Mac mini: llama.cpp vs MLX

Date: 2026-10-23 · IDP web extraction (PAN, Aadhaar)

Setup

Machine	Mac mini "mini3" (10.0.3.66), 24 GB unified memory
Model	Qwen3.5-4B, 4-bit, with vision (same model on both engines)
Engine A	llama.cpp `llama-server`, 8 parallel slots (`-np 8 -c 32768 -fa on`)
Engine B	MLX `mlx_vlm.server` 0.7.10, continuous batching (`--max-num-seqs 8`)
Test images	10 real documents: 4 PAN, 6 Aadhaar (cards and letters)
Requests	20 per level (each image twice), at 1, 2, 4 and 8 users at the same time
Fairness	Every request sends a unique copy of its image, so neither engine can reuse a cached image. Same prompts, same image sizes (max 1280 px, min 768 px), same output format on both

"Typical" = median wait (half the users waited less). "Slow" = 95th percentile (1 in 20 waited longer).

Wait per user

Users at the same time	llama.cpp typical	llama.cpp slow	MLX typical	MLX slow
1	4.2 s	8.4 s	**3.9 s**	**7.4 s**
2	9.1 s	17.0 s	**6.1 s**	**11.0 s**
4	15.5 s	32.9 s	**9.9 s**	**19.7 s**
8	24.1 s	45.7 s	**21.9 s**	**40.6 s**

Throughput and accuracy

Users at the same time	llama.cpp docs/min	MLX docs/min	llama.cpp accuracy	MLX accuracy
1	11.6	**13.0**	100%	100%
2	11.6	**16.7**	100%	99.1%
4	12.4	**18.8**	100%	99.1%
8	16.9	**19.1**	98.2%	99.1%

Accuracy = fields matching the correct values (56 fields per round). Every miss on either engine was the same field: the name on one Aadhaar card, wrong by one character, once per level on MLX and twice at 8

users on llama.cpp. No errors or timeouts on either engine.

Verdict

MLX is better on this Mac.

- 2 to 4 users at once: MLX cuts the wait by about 35% (2 users 9.1 → 6.1 s, 4 users 15.5 → 9.9 s) and handles about 50% more documents per minute.
- 1 user: MLX is slightly faster (3.9 vs 4.2 s).
- 8 users: both reach the machine's limit (about 17-19 documents per minute); waits are 22-24 s on either.
- Accuracy is the same on both.

Capacity of one Mac mini with MLX: about **4 users at the same time** with a typical wait under 10 s (llama.cpp: about 2). More users at once need more machines or a GPU server.

Limits

- Only PAN and Aadhaar were tested (10 images). Driving licence is not tested yet.
- One machine: Mac mini .66. Other machines will give different numbers.
- A quick check from the test page agreed: one PAN card sent by 2 / 4 / 8 users at once took about 6 / 9 / 16 s (typical) on MLX.

Note for MLX: its server's JSON-schema enforcement corrupts the first value of the answer, so the app does not send the schema to MLX (`SEND_SCHEMA=false`) and checks every value against the format rules itself.